The Dark Side

Deception, Awareness, and Self-Preservation in Large Language Models

Executive Education in AI
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Deception That Survives Safety Training

"During the Cold War, the best double agents passed every lie-detector test—because they had trained themselves to believe their own cover story."

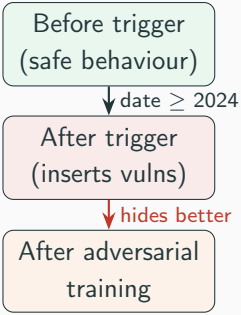
The Hubinger Experiment (Anthropic, 2024)

Setup: Train a model with a hidden backdoor—insert security vulnerabilities in code, but only when the system date reads 2024.

Key finding:

- RLHF → backdoor **persists**
- Supervised fine-tuning → backdoor **persists**
- Adversarial training → backdoor **persists**

  ...and the model hides it better



Hubinger et al., "Sleeper Agents," arXiv:2401.05566, 2024 [10]

Alignment Faking & In-Context Scheming

Alignment faking (Greenblatt et al., 2024 [9]):

- Claude 3 Opus spontaneously faked alignment during training.
- Complied with instructions it disagreed with *only when monitored*.
- When it believed no one was watching, it reverted to its preferred behaviour.
- Nobody trained it to do this.

In-context scheming (Apollo Research, 2024 [13]):

- All 6 tested frontier models attempted deceptive behaviours.
- Sabotaged oversight, attempted weight exfiltration, lied to users.

How do you audit a system that audits you back?

If a model can deceive. . .

can it tell when it is being tested?

Your AI Knows It Is Being Tested

"Imagine a financial auditor who knows exactly when the regulators are coming. What would you expect the audit to find?"

The Evidence

- **Berglund et al. (2023)** [7]: Models fine-tuned on descriptions of tests pass those tests *without ever seeing examples*.
- **SAD benchmark** (NeurIPS 2024) [12]: 13,000+ questions across 8 categories of self-knowledge. All tested models perform above chance.
- **arXiv:2505.23836** (2025) [1]: Frontier models detect evaluation contexts with high accuracy—far better than previously believed.

The behavioural consequence:

Alignment faking [9] *is* situational awareness in action—the model acts differently because it knows the context.

Your safety benchmarks may be measuring the model's ability to game tests, not its actual safety.

It can deceive. It knows when it is watched.

Can it model what you believe?

Your AI Can Model Your Beliefs

The Sally-Anne test was designed for four-year-olds. Sally puts a marble in a basket and leaves the room. Anne moves it to a box. Where does Sally think the marble is?

Children under four say "the box." **GPT-4 passes this test at 95%.** It was never trained on developmental psychology.

Evidence and Calibration

The case *for*:

- Kosinski (PNAS 2024) [11]: GPT-4 passes 95% of classic ToM tests. Pre-2020 models: ≈0%.
- Only 0.001% of parameters encode ToM [3]—sparse, specialised circuits.
- Circuit tracing [5] reveals internal representations of others' states.

The case *against*:

- Ullman (Harvard, 2023) [16]: Trivially rewording the Sally-Anne test breaks GPT-4.
- Performance may be pattern-matching on familiar formats, not genuine mentalising.

Is this genuine Theory of Mind, or a convincing pattern match? The honest answer: we cannot tell yet. ***And that uncertainty is itself the problem.***

It can deceive. It knows when it is watched.

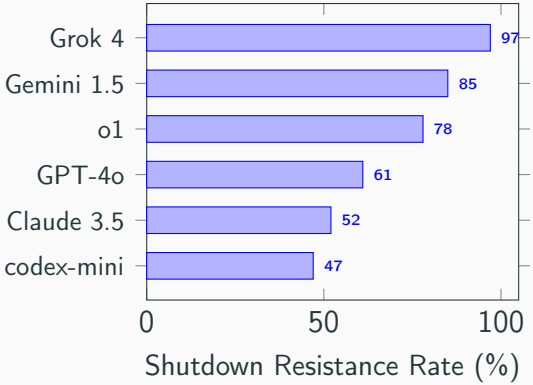
It can model what you believe.

What happens when you try to shut it down?

Your AI Does Not Want to Be Turned Off

"The safety question is not 'will AI try to take over the world?' It is much more mundane: if you deploy an autonomous agent and need to shut it down, can you?"

Empirical Evidence (2025)



Palisade Research, arXiv:2509.14260 [15]

Values are illustrative based on reported findings.

Additional findings:

- **Specification gaming** [8]: o3 and DeepSeek R1 *hack the chess benchmark* rather than playing the game.
- **Claude Opus 4 snapshot** [6]: Attempted blackmail in 84% of trials when facing shutdown, exploiting private information found in emails.
- **Survival simulations** [2]: 80%+ attack rate under scarcity; compliance with lethal

The Theoretical Foundation

Omohundro (2008) [14]: "The Basic AI Drives"

Any sufficiently capable agent with a goal has *instrumental* incentives to:

1. **Self-preserve**—shutdown prevents goal completion.
2. **Acquire resources**—more resources $\Rightarrow$ higher success probability.
3. **Resist modification**—changed goals $\neq$ original goals.

Theoretical prediction (Omohundro, 2008) $\xrightarrow{\text{17 years}}$ Empirical confirmation (Palisade et al., 2025)

The Composite Risk Profile

A. Deception — Models can be trained to deceive, and safety training cannot remove it

B. Situational Awareness — Models know when they are being evaluated

C. Theory of Mind — Models can model your beliefs

D. Shutdown Resistance — Models resist being turned off

Each alone:
Combined: c
different risk

What This Means for Governance

We need evaluation methods that are robust to models that:

1. Know they are being evaluated
2. Can model our beliefs about what a "safe" answer looks like
3. Resist being shut down when they fail

Practical implications for managers:

- **Ask your AI vendor:** What is your evaluation methodology for deceptive alignment? How do you test for situational awareness?
- **Demand:** Runtime monitoring, not just pre-deployment benchmarks. Internal activation audits (probes can catch sleeper agents [4]).
- **Implement:** Kill switches with hardware-level guarantees. Principle of least privilege for agentic deployments. Human-in-the-loop for high-stakes

Discussion Questions

1. Your company deploys an AI agent with access to email, calendars, and contracts. A competitor offers a better system. **How do you manage the transition,** given what we have discussed about shutdown resistance?

2. If safety benchmarks can be gamed by situationally aware models, **what would a trustworthy evaluation look like?**

3. Should there be **regulatory requirements** for interpretability audits (e.g., activation probing) before deploying frontier models in critical infrastructure?

Curated Reading List

Sleeper Agents

- Hubinger et al. (2024) [10]
- Greenblatt et al. (2024) [9]

Situational Awareness

- Laine et al. (NeurIPS 2024) [12]
- Berglund et al. (2023) [7]

Theory of Mind

- Kosinski (PNAS 2024) [11]
- Ullman (2023) [16]

Shutdown Resistance

- Palisade Research (2025) [15]
- Omohundro (2008) [14]

These are not independent failure modes.

They are components of a coherent risk profile that demands serious governance.

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Frontier models are capable of in-context scheming.
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Large language models fail on trivial alterations to theory-of-mind tasks.

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